Ranjan Sarkar

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Indian Institute of Technology Kharagpur

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ABOUT

PhD scholar at the Department of Artificial Intelligence, IIT Kharagpur, in the **RHPI (Receding Horizon and Predictive Intelligence) Lab** under the supervision of Dr. Prabhat Kumar Mishra. My research focuses on developing new robust RL algorithms with applications in LLMs and robotics.

RESEARCH INTERESTS

Machine Learning **Optimization** **Reinforcement Learning** **State-Space Models**

EDUCATION

PhD, Artificial Intelligence — **2025 – Present**
Indian Institute of Technology Kharagpur — CGPA 9.33/10

MSc, Mathematics — **2021 – 2023**
Indian Institute of Technology Kharagpur — CGPA 8.76/10

BSc, Mathematics — **2018 – 2021**
Ramakrishna Mission Vidyamandira (University of Calcutta) — CGPA 9.15/10

Class XII (WBCHSE, 2018) 88.6% · Class X (WBBSE, 2016) 94.57%
Raiganj Coronation High School (H.S.)

PUBLICATIONS

Ranjan Sarkar, Prabhat K. Mishra. "**LQG Solution for POMDP without estimating states: A minimum variance approach**." Accepted in *Indian Control Conference (ICC)*, 2027.

EXPERIENCE

Senior Business Analyst, EXL Service · Gurgaon, India — **2023 – 2024**

- Built data pipelines and analytics with SQL and SAS.
- Worked on business-as-usual (BAU) operations.

PROJECTS

Master's Thesis: Image Compression — **2022 – 2023**
Supervisor: Prof. Swanand Khare, Dept. of Mathematics, IIT Kharagpur

- Implemented tensor CP decomposition in Python.
- Applied the Alternating Least Squares (ALS) optimization method to obtain low-rank approximations of RGB image tensors.

TEACHING ASSISTANT

AI31201 · Reinforcement Learning	Autumn 2026
AI60209 · AI for Robot Autonomy	Autumn 2026
AI61006 · AI for Cyber Physical Systems	Spring 2026
AI60211 · Algorithmic and Mathematical Foundations for AI	Autumn 2025

SKILLS

Programming & Tools: Python, C, PyTorch, TensorFlow, Scikit-Learn, NumPy, Pandas, SymPy, CasADi, SQL, SAS, Jupyter, LaTeX

Coursework: Machine Learning, Deep Learning, Artificial Intelligence, Convex Optimization, Non-Linear Programming, Linear Algebra, Probability & Statistics, Real Analysis, Advanced Numerical Techniques, AI for Cyber Physical Systems, AI for Robot Autonomy

AWARDS

- Qualified Joint CSIR-UGC NET (June 2024) with All India Rank 83.
- INSPIRE Scholarship (2018–2023), DST, Govt. of India — awarded for top 1% in the WB HS board examination (2018).

INTERESTS

Photography, Singing, Swimming, Table Tennis